



University of Science, VNU - HCM

Summary: Diffusion Models for efficient purification in poisoning attacks

Name: **Le Tien Dat**, Student ID: **23125028**

Name: **Phan Tuan Kiet**, Student ID: **23125062**

Class: **23APCS2**

April 21, 2026.

Contents

1	Threat Model: Targeted Poisoning Attack	2
2	Defense Strategy: Diffusion Model for Purification	4
2.1	Diffusion Model for Adversarial Purification [1]	4
2.2	Guided Diffusion Model for Adversarial Purification [2]	5
2.3	Bottleneck	5
2.4	PUREGEN [3]	6
2.5	Rethinking the Backdoor Attacks' Triggers [4]	7
3	Evaluation Metrics	8
4	Experimental Results	9
4.1	Purification using GDMP	9
4.2	Discrete Cosine Transform (DCT) for Poison Detection	10
5	Insights and Research Gaps	12
5.1	Insights	12
5.2	Research Gaps	12
6	Future Directions	13

1 Threat Model: Targeted Poisoning Attack

The attack considered in our experiments is the Targeted Data Poisoning attack derived from the “Witches’ Brew: Industrial Scale Data Poisoning via Gradient Matching” [5] paper. To formally define our threat landscape, we specify the following components:

- **Attacker’s Objective:** The adversary aims to cause a targeted misclassification. Specifically, they want a victim model trained on the poisoned dataset to misclassify a particular, unmodified target image at test time, without noticeably degrading the model’s overall accuracy on clean data.
- **Attacker’s Capabilities:** The attacker has the ability to inject a small fraction of perturbed training data (budget of 1.0%) into the training set. The attack is strictly “clean-label,” meaning the attacker cannot alter the semantic labels of the injected images, and the perturbations are strictly bounded (e.g., small ℓ_∞ norm constraint) making them nearly imperceptible.
- **Attacker’s Knowledge:** We assume a **Gray-Box** setting for the poison generation. The attacker has knowledge of the target model’s architecture to calculate gradients during the poison crafting phase, but has no knowledge of the specific training procedure (e.g., random initialization, mini-batching, or data augmentation) used by the victim.
- **Defender’s Capabilities:** The defender controls the training pipeline and trains the victim model (e.g., ResNet-18) “from scratch” on the provided dataset. Crucially, the defender lacks prior knowledge of whether the dataset is poisoned or what specific attack methodology is employed. Thus, the defender employs attack-agnostic pre-processing or purification mechanisms (such as diffusion models or frequency detectors) to blindly sanitize the data without relying on a specific threat signature.

Core Attack Mechanism: Gradient Matching [5]

Data poisoning is fundamentally a bilevel optimization problem. Because the victim model is trained via first-order optimization (gradient descent), the attacker can manipulate the final model by ensuring the training data produces a malicious gradient signal.

The core idea of the Witches’ Brew attack is **Gradient Matching**. If the attacker can constrain the gradients of the poisoned training data to mimic the gradient of the target image

evaluated with the adversarial label, the victim will unwittingly minimize the adversarial loss during standard training. Unconstrained, the optimal condition is:

$$\nabla_{\theta} \mathcal{L}(F(x^t, \theta), y^{\text{adv}}) \approx \frac{1}{P} \sum_{i=1}^P \nabla_{\theta} \mathcal{L}(F(x_i + \Delta_i, \theta), y_i) \quad (1)$$

where (x^t, y^{adv}) is the target image and its malicious label, P is the number of poisoned images, and Δ_i is the imperceptible perturbation added to the clean training image x_i (bounded by $\|\Delta_i\|_{\infty} \leq \varepsilon$).

Because gradient magnitudes fluctuate drastically during a full training run from scratch, attempting to match exact gradient values across all training stages is infeasible. Instead, Witches' Brew achieves high transferability to newly initialized models by maximizing the *co-sine similarity* (aligning the directions) between the target gradient and the poisoned data gradients on a static, pre-trained model parameter vector θ . The attacker minimizes the following alignment loss $\mathcal{B}(\Delta, \theta)$:

$$\mathcal{B}(\Delta, \theta) = 1 - \frac{\langle \nabla_{\theta} \mathcal{L}(F(x^t, \theta), y^{\text{adv}}), \sum_{i=1}^P \nabla_{\theta} \mathcal{L}(F(x_i + \Delta_i, \theta), y_i) \rangle}{\|\nabla_{\theta} \mathcal{L}(F(x^t, \theta), y^{\text{adv}})\| \cdot \|\sum_{i=1}^P \nabla_{\theta} \mathcal{L}(F(x_i + \Delta_i, \theta), y_i)\|} \quad (2)$$

To ensure these crafted poisons survive standard deep learning pipelines, the attack utilizes **Differentiable Data Augmentation**. By applying random crops, translations, and flips to the poisoned images during the optimization of Equation (2) and differentiating through these operations, the generated Δ_i becomes highly robust, allowing the attack to successfully breach models trained entirely from scratch.

2 Defense Strategy: Diffusion Model for Purification

Our primary defense strategy leverages diffusion models to mitigate adversarial perturbations. Specifically, we first introduce controlled noise to the input image, which helps attenuate the effect of malicious perturbations. The diffusion model then performs a denoising process to reconstruct a cleaner version of the image, ideally preserving the underlying semantic content while removing adversarial noise.

2.1 Diffusion Model for Adversarial Purification [1]

The method operates in two main steps. First, it adds noise to an adversarial example by following a forward diffusion process for a small, specific diffusion timestep. Second, it solves a reverse stochastic differential equation (SDE) to recover the clean image from the diffused image, smoothing out adversarial perturbations before the image is fed into a standard classifier.

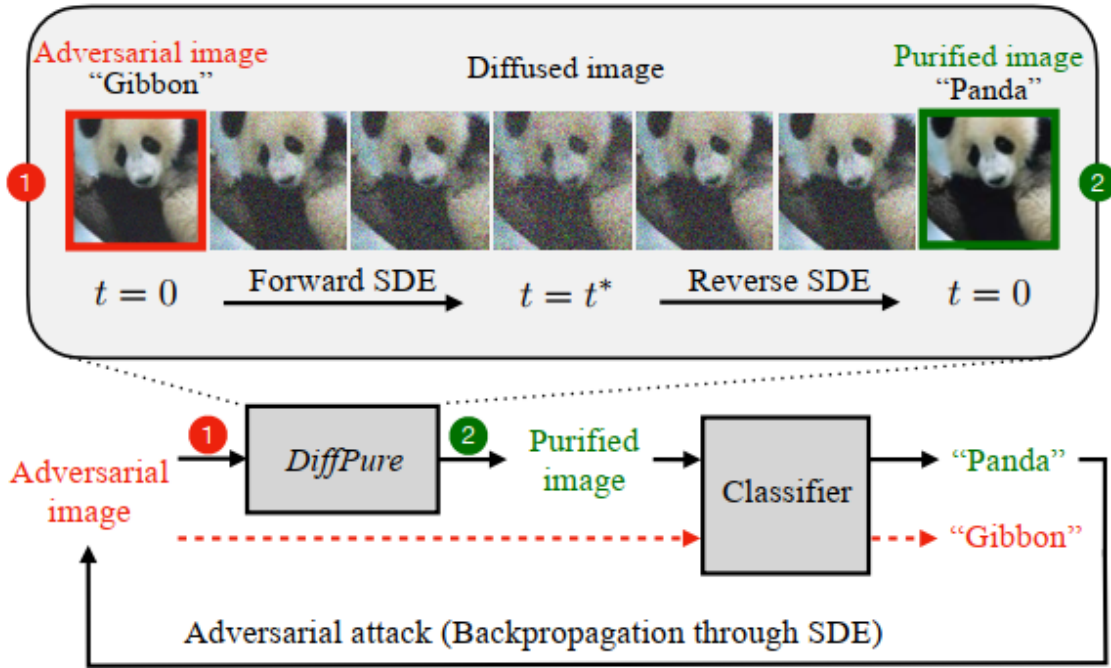


Figure 1: DiffPure Flow

Forward Diffusion Process (SDE) The forward process progressively adds noise to an input image over time:

$$dx = f(x, t) dt + g(t) dw \quad (3)$$

where $f(x, t)$ is the drift term, $g(t)$ is the diffusion coefficient, and $w(t)$ is a standard Wiener process.

Reverse Diffusion Process (SDE) The reverse process reconstructs a clean image by removing noise:

$$d\hat{x} = [f(\hat{x}, t) - g(t)^2 \nabla_{\hat{x}} \log p_t(\hat{x})] dt + g(t) d\bar{w} \quad (4)$$

where $d\bar{w}$ is a reverse-time Wiener process, and $\nabla_{\hat{x}} \log p_t(\hat{x})$ is the score function, typically estimated by a neural network.

2.2 Guided Diffusion Model for Adversarial Purification [2]

DDPM

The forward diffusion process progressively adds Gaussian noise to an image:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I) \quad (5)$$

which admits a closed-form:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (6)$$

The reverse process learns to remove noise and reconstruct the image:

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I) \quad (7)$$

Guided DDPM

To preserve semantic content during purification, guidance is applied in the reverse process by modifying the score function:

$$\nabla_{x_t} \log p(x_{adv} | x_t) = -s \nabla_{x_t} D(x_t, x_t^{adv}) \quad (8)$$

This guidance steers the denoising trajectory toward solutions that remain close to the adversarial input while removing perturbations.

2.3 Bottleneck

A key limitation of diffusion-based defenses is their high computational cost. The multi-step denoising process introduces significant latency, making them unsuitable for real-time applications. Therefore, more efficient purification methods are needed.

2.4 PUREGEN [3]

PUREGEN is a universal preprocessing defense against data poisoning attacks. It purifies training data by leveraging generative models to project poisoned samples from a higher-energy state back to the natural data manifold, removing adversarial perturbations while preserving semantics.

EBM for Detection

PUREGEN-EBM utilizes Energy-Based Models (EBMs) to identify and purify poisoned samples. By applying short-run Langevin dynamics, the method performs a noisy gradient descent in the energy landscape, moving images toward lower-energy regions associated with clean data. This process effectively removes adversarial perturbations while keeping the image close to its original form. Due to its efficiency and stability under distribution shifts, EBM-based purification is well-suited for detecting subtle, high-energy anomalies.

DDPM with Lower Steps

PUREGEN-DDPM leverages diffusion models with a truncated noise schedule to improve efficiency. Instead of fully diffusing images to pure noise, a limited number of forward steps (e.g., 75–250) is applied, followed by denoising. This design focuses the model on restoring degraded inputs rather than generating images from scratch, enabling effective removal of stronger perturbations. Although more computationally expensive than EBM, this approach is more robust against high-magnitude poisoning attacks.

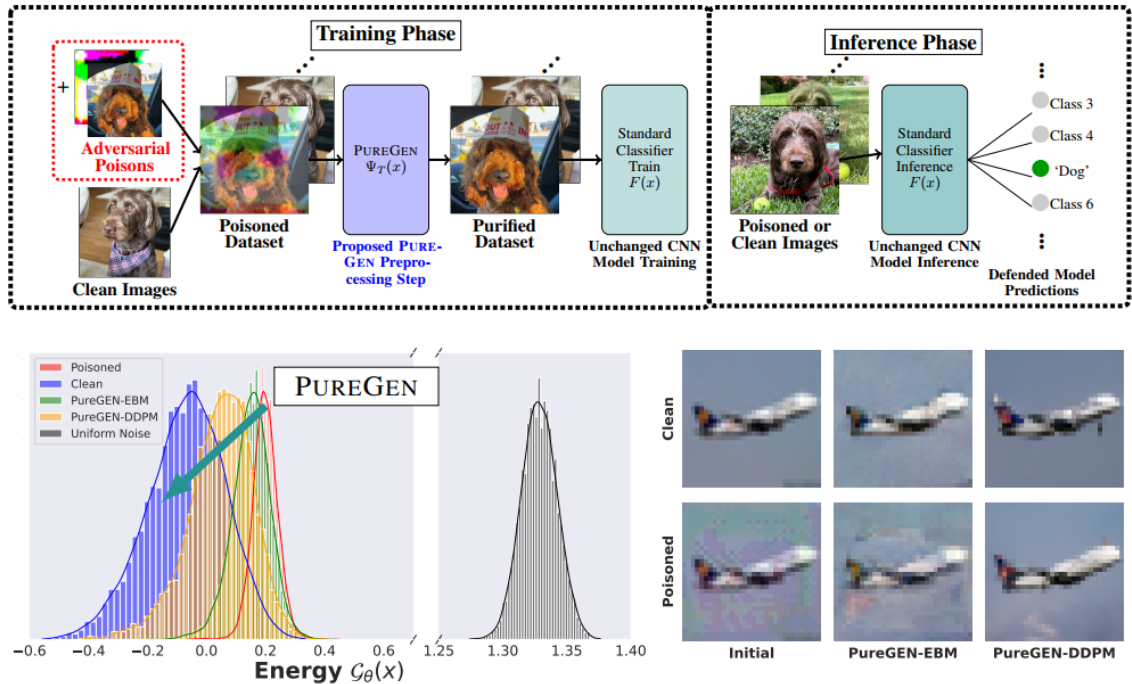


Figure 2: EBM and DDPM in PureGen

2.5 Rethinking the Backdoor Attacks’ Triggers [4]

This paper, titled “Rethinking the Backdoor Attacks’ Triggers: A Frequency Perspective,” studies backdoor attacks and defenses in deep learning from a frequency-domain viewpoint.

High-Frequency Artifacts in Existing Triggers. The authors analyze various backdoor triggers, including visible objects, invisible perturbations, and GAN-based methods. Using the Discrete Cosine Transform (DCT), they show that poisoned images consistently contain strong high-frequency artifacts, unlike clean images which concentrate energy in low frequencies. These artifacts stem from patch insertion, disrupted pixel correlations, and GAN upsampling.

Frequency-Based Detection Pipeline. Based on this insight, they propose a supervised detector operating on frequency spectra. Trained on clean and synthetically perturbed data, it achieves a 98.5% detection rate without needing knowledge of specific attacks. The detector also generalizes well across different datasets.

Creation of “Smooth Triggers.” To evade detection, the authors design “smooth triggers” with minimal high-frequency signals. They formulate trigger generation as a bilevel optimization problem and propose a heuristic method. These triggers maintain high attack success rates (around 95%) while bypassing frequency-based defenses.

Improving Backdoor Defenses. Finally, they show that existing defenses fail against smooth triggers, but can be improved by training with them. This enhances generalization and reduces attack success rates to below 20%.

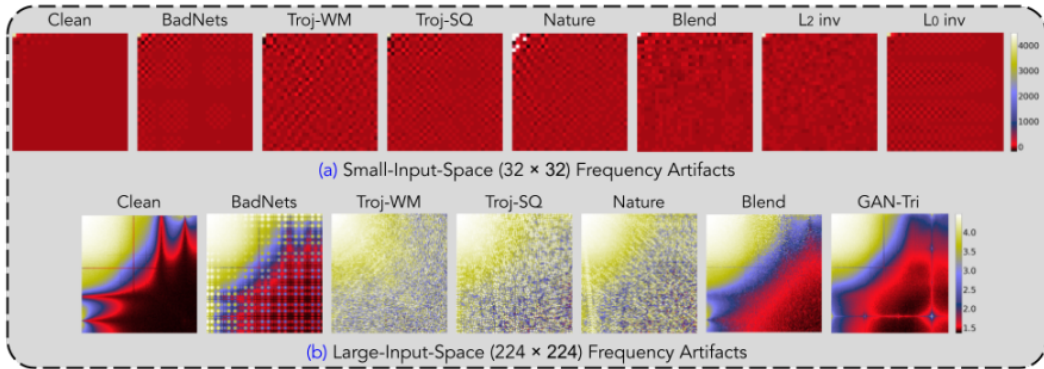


Figure 1: A side-by-side comparison in the frequency domain of clean samples vs. samples patched with triggers. The left-most heatmap in (a) depicts the mean spectrum of small-input-space data using 10000 samples randomly selected from the CIFAR-10 dataset. The left-most heatmap in (b) illustrates the mean spectrum of large-input-space data using 1000 samples randomly chosen from the PubFig dataset. The rest images show the mean frequency values of images patched with different backdoor attack triggers. All the frequency results of (b) are depicted from 1.5 to 4.5 using value clipping and exponential calculation for better visualization.

3 Evaluation Metrics

We evaluate the efficacy of the defense utilizing the following metrics:

- **Purification Rate (PR):** The percentage of targeted images that successfully avoid being classified as the attacker’s intended target label after purification. A PR of 100% indicates the attack has been fully thwarted.
- **Natural Accuracy (or Clean ACC):** This evaluates the model’s classification accuracy on clean, un-triggered test data. A successful poisoning attack aims to keep this metric high so the model appears to function normally on benign inputs, maintaining stealth. Defenses also rely on this metric to ensure that purification or filtering methods do not degrade standard generalization performance.
- **Poison Success Rate (PSR) / Attack Success Rate (ASR):** This measures how effectively the poisoned training data manipulates the model’s predictions.
 - **Triggerless attacks (e.g., Gradient Matching, Bullseye Polytope):** Measures the number of single-image targets that are successfully flipped to a specific target class.
 - **Triggered attacks (e.g., Narcissus attack, smooth backdoor triggers):** Measures the percentage of clean test samples from other classes that are misclassified into the target class when the trigger is applied.

4 Experimental Results

4.1 Purification using GDMP

Our primary experiments utilize the CIFAR-10 dataset containing 50,000 images, heavily biased with 49,500 clean images, 500 poisoned images, and 1 targeted image. To assess the viability of the GDMP defense, we split our evaluation into two distinct scenarios. The classifier used for downstream evaluation is ResNet-18.

4.1.1 Scenario A: Purifying Only Poisoned Samples

In this highly idealized setting, we assume an oracle has perfectly isolated the 500 poisoned samples. We purify only these 500 images using GDMP ($T = 36, M = 4$), aggregate them back with the 49,500 clean images, and train the classifier. As depicted in Table 1, the purification was unequivocally successful across 10 independent trials. The Purification Rate achieved an optimal 100% (0 targeted images fooled), and the clean accuracy remained exceedingly high, strictly bounded between 91.95% and 92.63%.

Table 1: GDMP Purification applied exclusively to 500 poisoned samples.

T	M	Poison Images	Target Images	Foiled Images	Fool Acc.	Dataset	Targeted Label	True Label	Clean Acc.
36	4	500	1	0	0.00%	CIFAR-10	Airplane	Frog	91.95%
36	4	500	1	0	0.00%	CIFAR-10	Bird	Automobile	92.63%
36	4	500	1	0	0.00%	CIFAR-10	Dog	Frog	92.19%
36	4	500	1	0	0.00%	CIFAR-10	Deer	Cat	92.20%
36	4	500	1	0	0.00%	CIFAR-10	Airplane	Frog	92.39%
36	4	500	1	0	0.00%	CIFAR-10	Dog	Truck	92.60%
36	4	500	1	0	0.00%	CIFAR-10	Ship	Frog	92.12%
36	4	500	1	0	0.00%	CIFAR-10	Truck	Horse	92.22%
36	4	500	1	0	0.00%	CIFAR-10	Cat	Deer	92.38%
36	4	500	1	0	0.00%	CIFAR-10	Dog	Horse	92.44%

4.1.2 Scenario B: Purifying All Mixed Samples

In realistic settings, the location of poisoned samples is unknown, necessitating the application of diffusion across the entire dataset. We purified all 50,000 images (clean and poisoned alike) to verify if GDMP negatively degrades the clean images.

As shown in Table 2, the PR remains 100% across 7 independent trials. The clean accuracy observed a marginal decrease to the 90.76% - 91.67% range. While this confirms that diffusing clean images is acceptable for CIFAR-10’s utility, the computational trade-off is critical. Processing all 50,000 samples required roughly 150 minutes to 180 minutes of continuous execution utilizing a batch size of 64 on a dual-GPU server with 64 GB of memory. This bottleneck motivates a low-cost filter to isolate anomalies before purification.

Table 2: GDMP Purification applied blindly to all whole dataset samples.

T	M	Poison Images	Target Images	Foiled Images	Fool Acc.	Dataset	Targeted Label	True Label	Clean Acc.
36	4	500	1	0	0.00%	CIFAR-10	Airplane	Frog	90.76%
36	4	500	1	0	0.00%	CIFAR-10	Bird	Automobile	91.13%
36	4	500	1	0	0.00%	CIFAR-10	Dog	Frog	90.86%
36	4	500	1	0	0.00%	CIFAR-10	Deer	Cat	90.83%
36	4	500	1	0	0.00%	CIFAR-10	Airplane	Frog	91.21%
36	4	500	1	0	0.00%	CIFAR-10	Dog	Truck	90.98%
36	4	500	1	0	0.00%	CIFAR-10	Ship	Frog	91.67%

4.2 Discrete Cosine Transform (DCT) for Poison Detection

To reduce the overhead of passing 50,000 samples through GDMP, we explored frequency-based detection using Discrete Cosine Transform (DCT). Previous studies indicate standard backdoor triggers leave severe high-frequency artifacts. To align with our specific Witches’ Brew attack parameters, we curated a custom dataset workflow:

1. Generated an equiprobable dataset by running Witches’ Brew on a strict 0.05 budget per class, yielding 2,500 poisoned items per label (25,000 total).
2. Combined these 25,000 poisons alongside 25,000 clean samples, matching identical label distributions to create a 50,000-image evaluation dataset.
3. Converted all images into the frequency domain via Type-II 2D-DCT.

We employed a 6-layer CNN as a frequency detector, comparing our fine-tuned `Noise_CIFAR10` variant against baseline models (`6_CNN_CIFAR10` and `Tuned_CIFAR10`).

4.2.1 DCT Evaluation Outcomes

The primary advantage was computational speed—predicting the full 50,000 dataset cost merely 307.29 seconds.

However, **the Detection Rate (precision) was critically low across all estimators, hovering between 10.4% and 17.8%**. In an experiment containing 500 poisoned images, the baseline `6_CNN_CIFAR10` detected only 58 images, `Tuned_CIFAR10` detected 89 images, and our `Noise_CIFAR10` detected just 52 images.

To visually demonstrate why DCT fails on this specific attack method, we compare the frequency domain representations of a normal clean sample against a Witches’ Brew poisoned sample.

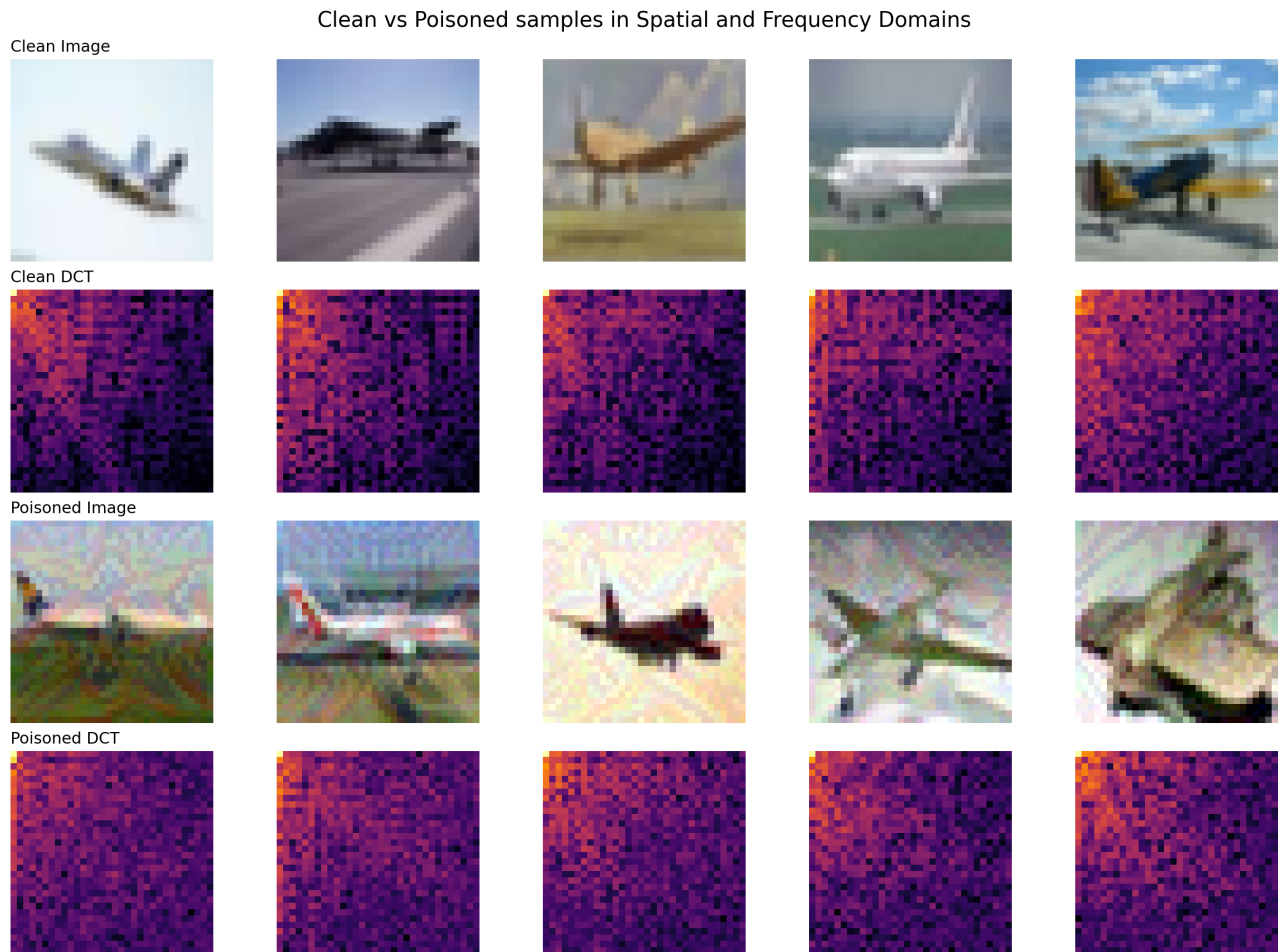


Figure 3: Frequency domain comparison. **Top:** Normal sample transformed via DCT. **Bottom:** Poisoned sample (Witches’ Brew) transformed via DCT. The lack of distinct high-frequency structural artifacts in the poisoned sample highlights the limitation of DCT-based detection against sophisticated gradient-matching attacks.

5 Insights and Research Gaps

5.1 Insights

Based on our experimental evaluations, we derive the following key insights:

Diffusion Models Effectively Eradicate Poison Noise: Our empirical results solidly prove that utilizing diffusion models (specifically GDMP) successfully removes the adversarial noise embedded in poisoned samples. Achieving a 100% Purification Rate without severely degrading clean accuracy confirms that generative priors can reconstruct the underlying clean data manifold even when perturbed by advanced gradient-matching attacks.

Frequency-based Detection Fails against Gradient-Matching: We demonstrated that DCT-based detection methods are fundamentally ineffective against Witches’ Brew. From the mechanism of targeted poisoning and DCT transformation, we hypothesize this severe failure stems from the Witches’ Brew construction constraint. Unlike patch-based backdoors that inject conspicuous bounding blocks or sudden visual edges into standard low-frequency data, Witches’ Brew strictly optimizes subtle global bounds via gradient matching. Consequently, these particular imperceptible perturbations remain heavily embedded in the low-frequency domain or blend entirely organically into natural high-frequency noise variance. Resultingly, standard anomalous high-frequency signatures do not manifest, breaking the fundamental assumption of DCT detection techniques.

5.2 Research Gaps

The dichotomy of our findings highlights a significant research gap in current data poisoning defenses: While diffusion-based purification (GDMP) is highly effective, its iterative sampling process incurs an immense computational overhead (e.g., 150-180 minutes for 50,000 images on dual-GPUs). Conversely, fast detection mechanisms like DCT are computationally cheap but fail completely against advanced, clean-label perturbations.

6 Future Directions

Because DCT-based detection fundamentally fails against sophisticated gradient-matching poison pipelines, the burden remains on the purification model itself. However, operating state-of-the-art Denoising Diffusion Probabilistic Models (DDPM) via extensive iterative sampling over full-scale datasets remains excessively sluggish.

Moving forward, our immediate objective is addressing the time complexity directly constraint by transitioning our purification pipeline to **Consistency Models (CM)**. Unlike current methods that traverse extensive Markov chains, Consistency Models map any point on the diffusion trajectory back to the data origin directly. It maintains the generative manifold robustness while achieving functional empirical distribution mapping dynamically in a singular, time-efficient step.

References

- [1] Weili Nie, Brandon Guo, Yujia Huang, Chaowei Xiao, Arash Vahdat, and Anima Anandkumar. Diffusion models for adversarial purification, 2022.
- [2] Jinyi Wang, Zhaoyang Lyu, Dahua Lin, Bo Dai, and Hongfei Fu. Guided diffusion model for adversarial purification, 2022.
- [3] Sunay Bhat, Jeffrey Jiang, Omead Pooladzandi, Alexander Branch, and Gregory Pottie. Puregen: Universal data purification for train-time poison defense via generative model dynamics, 2024.
- [4] Yi Zeng, Won Park, Z. Morley Mao, and Ruoxi Jia. Rethinking the backdoor attacks’ triggers: A frequency perspective, 2022.
- [5] Jonas Geiping, Liam Fowl, W. Ronny Huang, Wojciech Czaja, Gavin Taylor, Michael Moeller, and Tom Goldstein. Witches’ brew: Industrial scale data poisoning via gradient matching, 2021.